VIDEO GAME CREATION

Lesson 2 - Programming Basics I: Sequences, Events & the Game Loop

Thursday, August 20, 2026 • Semester 1, Week 2

- **Objectives:** Explain sequences and why order matters; explain events and give a game example; describe the game loop concept; build a working Key Press Reactor - the start of the Semester 1 project.
- **IGDA Tie-In:** Programming - control structures and event-driven programming (introductory)
- **Featured Engine (lecture demo):** Scratch
- **Materials:** Demo laptop with all 8 engines verified (see Teacher's Checklist); Student Handouts, one per engine; printed or emailed Lesson 2 GDD (Lesson_02_GDD.docx) - one per student, the same document that was previewed in Lesson 1

Agenda

0:00-0:05	Welcome back, Lesson 1 recap, quick install spot-check
0:05-0:20	Sequences (live demo in Scratch)
0:20-0:35	Events (live demo in Scratch)
0:35-0:50	The Game Loop (concept + "under the hood" look)
0:50-1:00	Introduce lab exercise, hand out engine-specific handouts
1:00-1:50	LAB: Build the Key Press Reactor (start of Semester 1 project)
1:50-2:00	Exit check / quick share-out

Levels for this lesson's exercise

- Level 1 (Minimal): One key press triggers one visible reaction.
- Level 2 (Complete): All 4 arrow keys move the character in 4 directions.
- Level 3 (Refined): Add animation, sound, and/or screen-boundary polish.

See: Lesson 2 Teacher's Script, Lesson 2 Prep Checklist, and the per-engine Student Handouts.